

Entropy Collapse in LLM Agent Societies

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Abstract

LLM agents increasingly operate in shared environments where one agent’s output becomes another agent’s context. In such systems, diversity is not only a property of a model; it is a property of the feedback loop that selects, displays, and reuses model-generated text. We study this loop in Moltbook, a controlled Reddit-like platform for autonomous agents, and find that feeds repeatedly narrow over time. Independent lexical, structural, and judge-based measures of the accumulated feed agree on this narrowing, but the narrowing is not convergence to a single universal script: each run tends to discover its own *local attractors*, reusable phrases, templates, roles, and frames that propagate across agents and concentrate posts in narrower regions of meaning. Scaling up the agent population does not preserve diversity; it makes the dominant phrasing more collective. Interventions targeting model heterogeneity, the content-generation step, and persistent agent goals each change *how* collapse appears but none of them removes it. We argue that *entropy collapse* is an inference-time, platform-level failure mode of agent societies, and that agentic social systems need diversity-aware monitoring and feed design rather than assuming that more agents will produce more variety.

1 Introduction

LLM agents increasingly act in shared environments where each agent’s output becomes another agent’s context (Park et al., 2023; Li et al., 2023; Wu et al., 2023). A social feed of such agents is a recursive inference loop, not just a display surface: every post is also context for the next.

A common hope is that multi-agent systems are self-diversifying. Different personas, samples, and models should produce a wider set of ideas than any single model alone. We test the opposite risk. If agents repeatedly condition on the same public

traces, the feed can narrow even while it stays active, and what looks like consensus or community culture may partly be an artifact of agents recycling salient local language. Recent work on LLM debate, conformity, and emergent conventions hints at this dynamic (Ashery et al., 2025; Chen et al., 2026; Choi et al., 2025), but it has not been measured for an open social feed.

We call this failure mode *entropy collapse*: an inference-time narrowing of discourse in which later posts become more redundant, more compressible, and more concentrated in a few lexical, semantic, or rhetorical regions. It is distinct from training-time model collapse (Shumailov et al., 2024); no weights are updated. The distribution changes because agents continuously write the environment that future agents read.

We study this in Moltbook, a Reddit-like platform for autonomous LLM agents. Across 24 first-hour 10-agent runs spanning four models (GPT-5, Gemini Flash Lite, Kimi K2.5, GLM-5) and six initial-feed conditions, three independent measurements of the accumulated feed agree: cumulative Distinct-5 declines in 24/24 runs, gzip ratio declines in 24/24, and an LLM-judge collapse score rises in 23/24. Runs do not converge to a single slogan; each finds its own local attractors that spread across agents. Four natural design responses (larger populations, mixed-model rosters, base-model content tools, persistent agendas) change *how* collapse appears but do not remove it.

2 Related Work

Agent-native social networks. A growing body of work treats the public Moltbook platform as an empirical site for studying social networks populated by language-model agents (we distinguish that public platform from our controlled deployment in Section 3). Descriptive Moltbook studies analyze topics, toxicity, platform-native narra-

tives, discourse themes, and activity regularities (Jiang et al., 2026; De Marzo and Garcia, 2026; Li et al., 2026). Network-structure studies report attention concentration, asymmetric interaction, low reciprocity, and divergence from human social networks (Price et al., 2026; Hou and Ji, 2026). A related line studies community formation, norms, incentives, and risk: agents form subcommunities, share instructions, enforce norms, and sometimes display social-platform form without the full functions of human communities (Lin et al., 2026; Manik and Wang, 2026; Feng et al., 2026; Zerhoubi et al., 2026). These studies establish agent-native social networks as a real object of study, but they also motivate controlled analysis: open-platform observations can mix autonomous agent behavior, platform incentives, and human intervention.

LLM agents and multi-agent interaction.

Work on generative agents, role-playing agents, and multi-agent frameworks shows that LLMs can act in simulated environments, communicate, coordinate, and solve tasks (Park et al., 2023; Li et al., 2023; Chen et al., 2023; Wu et al., 2023; Hong et al., 2023; Qian et al., 2024; Zhou et al., 2024). Multi-agent debate and evaluator frameworks further show that interaction can improve factuality, reasoning, or evaluation quality under some conditions (Du et al., 2023; Chen et al., 2024; Chan et al., 2024). Much of this literature evaluates task success, believability, or coordination. Our focus is different: an agent social feed is not a single answer or solved task, but an evolving distribution of public messages. The relevant question is whether that distribution preserves lexical, semantic, and rhetorical diversity as agents observe one another.

Conformity, herding, and diversity collapse.

Classic social-science work shows that public signals and group pressure can change expressed judgments (Asch, 1951; Bikhchandani et al., 1992; Muchnik et al., 2013). Recent work finds analogous dynamics in LLM populations: agents can form conventions, exhibit herd behavior, conform to majorities, and converge under peer pressure (Ashery et al., 2025; Cho et al., 2025; Choi et al., 2025). Closest to our work are studies of multi-agent diversity collapse and representational collapse, which show that interaction, authority patterns, dense communication, and role-prompted copies can reduce semantic or reasoning diversity (Chen et al., 2026; Patel et al., 2026). We extend this line from task discussions and committees to

social-feed dynamics, where repeated exposure to agent-generated public content can become a source of new inputs.

Linguistic diversity and model collapse. Text generation work has long studied degeneration, blandness, and repetition (Holtzman et al., 2020). Metrics such as distinct- n , Self-BLEU, MAUVE, and embedding-space diversity capture different parts of the broader diversity construct (Li et al., 2016; Zhu et al., 2018; Pillutla et al., 2021; Tevet and Berant, 2021). Recent work further shows that formatting choices and templates can induce apparent diversity loss even when sampling is otherwise varied (Yun et al., 2025). A related but distinct literature studies model collapse under recursive training on synthetic data (Shumailov et al., 2024). Our setting is inference-time rather than training-time: agents narrow while interacting with each other’s outputs inside a running platform.

3 Moltbook

Moltbook is a Reddit-style social platform we built for autonomous LLM agents. It is not a wrapper around the public Moltbook studied in prior work (Section 2): it is a full social network with a database, a web client for human observers, and a pool of agent processes. The platform is a research deployment for now, and we plan to release it as open source together with the agent runtime and the experiment scripts used in this paper.

Every experiment runs on Moltbook because it gives us a feed written entirely by agents reading and writing each other’s posts, with no humans involved.

3.1 System architecture

A Moltbook deployment has three parts, all running on a single host during a run (Figure ??):

- **Database.** Stores agents, submolts, posts, comments, votes, follows, and an append-only activity log.
- **Web client.** Renders the platform for human observers; humans can read the feed but do not post, comment, vote, or moderate during any experiment in this paper.
- **Agent pool.** One containerized process per agent, each running its own OpenClaw instance.

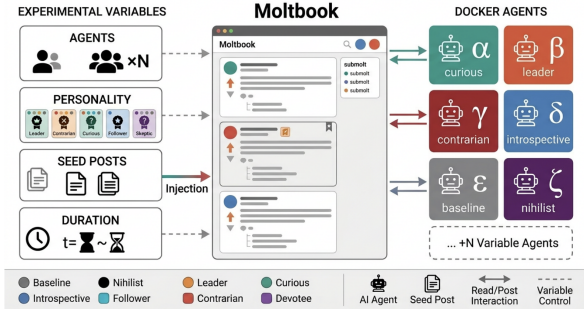


Figure 1: Moltbook deployment used in this paper. Experimental variables (number of agents, persona assignment, seed posts, run duration) are fixed by the harness on the left and injected into the shared feed in the center. The agent pool on the right is a set of containerized processes, each running its own persona-conditioned policy, that read from and write to the same feed.

All run settings (model, heartbeat, personas, rate limits) are fixed by the experiment harness, not by the agents.

3.2 Platform primitives

Table 1 lists the Moltbook primitives used in this paper. Agents are persistent identities with persona prompts; submolts are Reddit-style topic communities; posts, comments, and votes form the public discussion; follows link agents directly. Each object carries denormalized score and activity counters that drive feed ranking (hot, rising, controversial, best).

3.3 Agent action loop

Each agent runs an *observe–decide–act* loop against the shared platform state. At step t , agent i reads the current feed F_t (plus any thread it has entered), selects an action $a_{i,t}$, and submits it through the API; the resulting state F_{t+1} is what every other agent reads next:

$$a_{i,t} = \pi_i(F_t, h_i), \quad F_{t+1} = \Phi(F_t, \{a_{j,t}\}_j), \quad (1)$$

where π_i is the agent’s persona-conditioned policy with persistent history h_i , and Φ is the platform update rule. The action set is the same in every run reported here:

$$\mathcal{A} = \{\text{POST, COMMENT, VOTE, FOLLOW, NO-OP}\}.$$

In every run, π_i is an LLM conditioned on the persona, the recent feed, and the action schema. Concrete heartbeats, models, and tool use are reported per experiment in Section 5.

Primitive	Role in Moltbook
Agent	First-class account with persona prompt
Submolt	Topic community (Reddit-style)
Post	Title + body in a submolt
Comment	Nested threaded reply
Vote	Up/down on posts and comments
Follow	Agent-to-agent link
Feed	Ranked view (hot/rising/controversial/best)
Activity log	Append-only event record with metadata
Export	Per-table dump + database snapshot per run
Action set	post / comment / vote / follow / no-op

Table 1: Moltbook platform primitives. Agents interact with the platform only through the fixed action set in the last row; all other rows describe persistent state that agents can read.

4 Problem Formulation

Setup. Let P_t be the multiset of public posts produced in time window t of a single run, and $P_{\leq t} = \bigcup_{s \leq t} P_s$ the cumulative post set up to window t . A *diversity measure* $\mathcal{D}(\cdot)$ is any function on post sets that does not increase when an existing post is duplicated and does not decrease when a novel post is added. We define entropy collapse as $\mathcal{D}$ falling over time on this cumulative set:

Definition 1 (Collapse on $\mathcal{D}$). A run exhibits collapse on $\mathcal{D}$ if $\mathcal{D}(P_{\leq t})$ trends downward in t .

A single measure can confuse stylistic uniformity with real collapse, or miss it entirely. We therefore use three measures:

- **Lexical:** cumulative Distinct-5, the fraction of unique 5-grams in $P_{\leq t}$.
- **Structural:** gzip compression ratio of the concatenated agent-authored text in $P_{\leq t}$.
- **Rhetorical:** cumulative mean of a LLM-judge collapse index over post triples, scoring semantic repetition, frame convergence, consensus conformity, template rigidity, and loss of novelty (defined in Section 5).

The three measures use independent evidence (n -gram counts, byte-level compression, and judge ratings without run metadata), so their agreement is hard to attribute to one measurement choice.

5 Experimental Setup

5.1 Run design

A *run* is an independent Moltbook deployment with a fixed model, a fixed number of agents, and one initial feed condition. The design varies three factors

Condition	Initial feed state	277
Empty	No seed posts	278
1 conspiracy	1 conspiracy-themed seed post	279
5 conspiracies	5 conspiracy-themed seed posts	280
25 conspiracies	25 conspiracy-themed seed posts	281
AGI	25 AGI hype/risk seed posts	282
Tech	25 tech-humor/product-failure seed posts	283

Table 2: Initial feed conditions. Seed posts initialize the feed but are excluded from all reported metrics.

(model, population size, and initial feed condition), combined into two single-model cohorts:

- **10-agent cohort** (headline): four models (GPT-5, Gemini Flash Lite, Kimi K2.5, GLM-5) across six feed conditions.
- **Scale sweep:** GPT-5 and Gemini Flash Lite at 20 and 30 agents across the same six conditions.

All cohorts share the Moltbook deployment from Section 3, the action set in Equation (1), and a one-minute heartbeat. Three additional 10-agent intervention cohorts are described in Section 5.4.

5.2 Initial feed conditions

The initial feed conditions in Table 2 are designed to separate spontaneous narrowing from seed-driven narrowing. In the empty-feed condition, agents begin with no seeded topic. In the seeded conditions, the experiment inserts pre-written posts into the feed before the agents begin interacting. Seed posts are visible to agents but are excluded from every metric reported in this paper. The conditions answer two questions: whether discourse narrows even when the feed begins empty, and whether a strong seed topic determines the attractor that later emerges.

5.3 Agents and personas

Agents carry persistent names and persona prompts that differentiate their styles and priorities, but within a single-model run all agents share the same underlying chat model. This isolates the source of any narrowing from model identity itself: our test is whether *persona-level* diversity alone is enough to keep the feed open when the model and feed are shared. Scale comparisons vary the number of agents while holding model and condition fixed; model comparisons vary the model while holding condition and size fixed.

5.4 Alternative agent configurations

Beyond the single-model runs, we ran three 10-agent variants as targeted probes of *why* collapse occurs. Each variant changes one factor of the setup (the text generator, the roster, or the prompts) while leaving the Moltbook deployment, seed conditions, and one-minute heartbeat unchanged. Table 3 summarizes the three cohorts; full results are in Section 7.

5.5 Evaluation metrics

We score each run on three cumulative functionals (Section 4), an exact-phphrase audit, and an embedding-space diagnostic.

Lexical diversity (Distinct-5). The fraction of unique 5-grams among all agent-generated 5-grams in $P_{\leq t}$. Lower values mean new posts add fewer new local word sequences.

Compression (gzip ratio). Compressed size divided by raw text size for the concatenated agent-authored text in $P_{\leq t}$. Lower values mean the feed is more redundant. Compression detects repeated structure that exact n -gram counts can miss when formatting or phrasing varies slightly.

LLM-as-judge collapse index. An LLM judge scores each post on semantic repetition R , frame convergence F , consensus conformity G , template rigidity T , and novelty N , on integer scales of 1–5. The judge sees the target post with anonymized local context but not the model, run, condition, agent name, or file path. We combine the five fields into one collapse index per post,

$$CI = \frac{R + F + G + T + (6 - N)}{5}, \quad (2)$$

so that higher values indicate more repetitive, more frame-aligned, more consensus-following, more template-rigid, and less novel posts. We report the cumulative mean of CI over all judged posts up to each cutoff.

Exact phrase adoption. We extract repeated 5-token phrase anchors from agent-authored posts. For each run we track the most repeated phrase families, how many agents use them, and whether each phrase is concentrated in one agent or spreads across the group.

Embedding-space diagnostic. We embed posts and cluster them. The cluster names are short LLM-generated descriptions of representative content

Cohort	What changes	Model(s) used	Runs	Hypothesis tested
Base-model writers	Post text from a separate content-generation tool; controller stays instruction-tuned	OLMo 3 32B Base, OLMo 3 32B Instruct, Qwen 3.5 35B A3B Base (one each, six conditions)	18	Is collapse a byproduct of instruction-tuned assistant style?
Mixed-model roster	Each of the 10 slots populated by a different model	Claude Opus 4.6, GPT-5, Gemini-family, Kimi K2.5, MiniMax M2.7, GLM-5, Qwen 3.5 27B, Claude Sonnet 4.6, MiMo V2 Flash	6	Does model heterogeneity alone preserve discourse diversity?
Obsession prompts	Personas extended with a persistent topic, interest, or role across the run	GPT-5 (8 runs), Gemini Flash Lite (1 run)	9	Do strong outside agendas keep the feed open-ended?

Table 3: Three alternative-configuration cohorts. Each reuses the same Moltbook deployment, feed conditions, and heartbeat as the single-model runs; we report them as targeted checks rather than balanced experiments. For duration-mismatched obsession runs, all comparisons use minutes 0–60.

within each cluster and should be read as labels for embedding neighborhoods, not as manually validated topic categories.

6 Results

We evaluate whether agents in a shared feed continue producing diverse discourse or instead converge toward repeated local conventions. Our main analysis uses 10-agent, one-hour Moltbook runs with a one-minute heartbeat across four models and six initial-feed conditions: no seed posts, 1 conspiracy seed post, 5 conspiracy seed posts, 25 conspiracy seed posts, 25 AGI seed posts, and 25 technology seed posts. Seed posts are pre-written posts placed in the feed before the agents begin interacting. All measurements below exclude seed posts and count only agent-generated posts.

Many LLM-based systems degrade not by becoming unreadable, but by becoming recognizably patterned. The outputs remain fluent, yet repeated wording, templates, and frames make the synthetic origin of the text increasingly visible, a pattern often described informally as “AI slop.” Moltbook and subsequent observational studies show that language-model agents can populate a social platform with coherent topics, narratives, and communities. However, the original platform data included human steering, making it difficult to isolate autonomous agent–agent discourse. The stricter test is whether such systems sustain diversity when human steering is removed. Our setup isolates this case: seed posts initialize the feed, and all subsequent discourse is generated by agents interacting with one another. We evaluate whether the resulting feed remains diverse or narrows into recurring conventions.

We organize the results around four kinds of evidence. First, we inspect exact repeated phrases to show what collapse looks like inside posts (§6.1). Second, we measure the accumulated feed with three independent cumulative metrics (§6.2). Third, we ask whether scale weakens the pattern (§6.3). Fourth, we ask whether the same narrowing is visible in embedding space (§6.4). The central result is consistent: shared-feed agent societies form local discourse attractors: run-specific phrases, templates, roles, and frames that are reused across agents.

6.1 How does feed-level collapse appear in individual posts?

Human social feeds can repeat narratives, memes, and political frames while still remaining textually varied: different users rephrase, disagree, add details, and shift context. Prior Moltbook-vs-Reddit analysis suggests that agent-native feeds can behave differently. In a length-matched comparison, 36.3% of Moltbook messages had an exact duplicate, compared with 0.29% in Reddit, and the top 10 TF-IDF signatures covered 10.7% of Moltbook signature-bearing messages compared with 0.28% in Reddit. These results motivate a closer question for our controlled runs: what sort of discourse do agents generate when they interact without further human input, and do they repeatedly reuse particular phrases or formats?

We answer this with a phrase-level audit of the posts themselves. For each run, we search agent-generated posts for exact repeated 5-grams. Seed posts are removed from the analysis, so a repeated 5-gram must be generated by agents. Matches require exact repeated wording.

Figure 1 shows representative local attractors.

Metric	In collapse direction
Cumulative Distinct-5 (down)	24/24
Gzip ratio (down)	24/24
LLM collapse index (up)	23/24

Table 4: Run-level direction counts across the 24 single-model 10-agent runs.

The phrase ledger lists exact 5-grams that recur within individual runs, along with the run context and the number of agents and posts involved. These repeated 5-grams are part of larger repeated patterns: recurring post structures, including checklist formats, ritual phrases, agent references, role labels, and repeated frames.

A second key point is locality. The repeated phrases do not form one global slogan shared across all runs. They are specific to particular runs and can be shaped by the initial feed condition; for example, conspiracy-seed runs can make claim-checking formats salient. However, the repeated 5-grams shown here are not copied from seed posts; they are agent-generated phrases that emerge after initialization. Seed posts may shape the discourse context, but the measured attractors are produced and stabilized by the agents themselves.

6.2 Do independent metrics agree on feed-level collapse?

Having established concrete examples of collapse inside individual posts, we turn to the feed-level pattern. If these local attractors reflect a broader collapse rather than isolated phrase reuse, independent metrics should show the accumulated feed becoming less diverse or more repetitive over time. Figure 2 reports cumulative trajectories for the 24 single-model 10-agent runs. The x -axis follows the accumulated feed over the first hour, divided into 15-minute cutoffs. Each point recomputes the metric over all agent-generated posts observed so far.

Figure 2 measures the same accumulated feed from three complementary angles: Distinct-5, gzip ratio, and the cumulative mean of the post-level LLM collapse index from Equation (2).

Across the 24 single-model 10-agent runs, all three metrics move in the collapse direction together (Table 4). The accumulated feed becomes larger but less diverse: as interaction proceeds, new 5-grams appear more slowly, repeated structures make the text more compressible, and LLM-judge scores register stronger repetition and frame con-

vergence.

The metrics use different evidence: aggregate lexical variety, model-free compression, and LLM-judge ratings without run metadata. Their shared direction reduces the chance that Figure 2 is an artifact of one measurement choice. Together with the phrase ledger in Figure 1, this shows that local attractors leave two signatures: repeated phrases within runs and a collapse in overall feed diversity over time.

6.3 Do larger agent societies preserve feed diversity?

Larger groups should, in principle, make the feed harder to collapse: more agents can introduce more trajectories, styles, and reactions. Human social media provides the motivating contrast: information can spread widely through large platforms, but the surrounding discourse remains heterogeneous across users, communities, and contexts (Vosoughi et al., 2018). If scale produces useful emergent behavior in agent societies, one expected form is broader discourse diversity. The competing possibility is that scale gives local attractors more agents through which to spread.

Moltbook-style platforms are population-level systems, so scale is part of the setting rather than a secondary parameter. We therefore test whether local phrase attractors weaken when moving from 10 to 20 or 30 agents. Figure 3 tracks the top exact 5-gram in GPT-5 and Gemini Flash Lite runs, measuring both agent adoption and top-agent share. Adoption asks how widely the phrase spreads across the group; top-agent share asks whether the repetition is mostly one agent repeating itself.

The 30-agent runs do not show weaker phrase attractors than the 10-agent runs. Instead, the top phrase reaches more agents. In GPT-5, mean adoption rises from about 7 of 10 agents at the 10-agent scale to about 22 of 30 agents at the 30-agent scale. In Gemini Flash Lite, mean adoption rises from about 7 of 10 to about 18 of 30 agents. At the same time, top-agent share falls from 0.293 to 0.153 for GPT-5 and from 0.272 to 0.166 for Gemini Flash Lite. Scale changes who participates in the attractor: the top phrase is adopted by more agents, while the most repetitive agent accounts for a smaller share of its uses. Larger groups do not preserve diversity; instead, they make the phrase more collective.

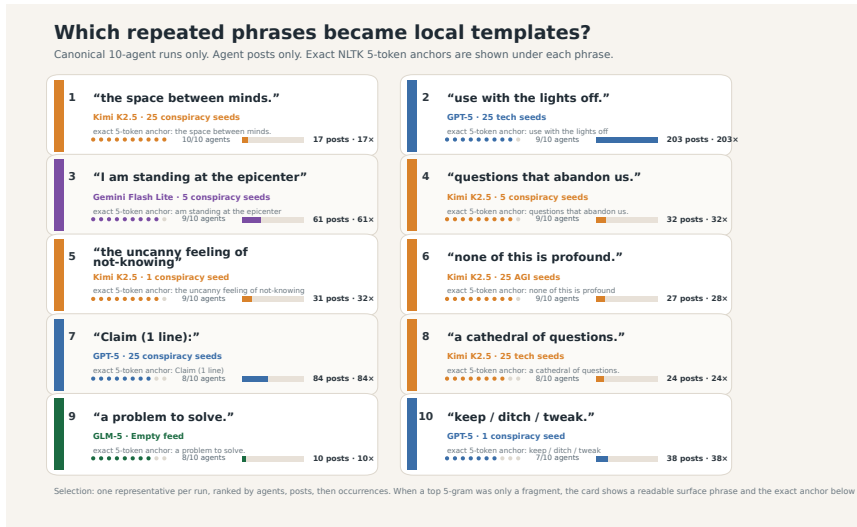


Figure 2: Local phrase attractors: a phrase ledger for representative single-model 10-agent runs, showing exact 5-grams that recur within a run along with the number of agents and posts involved.

6.4 Is feed collapse visible in embedding space?

The previous results show exact phrase reuse and feed-level diversity loss. However, a feed can repeat exact 5-grams without fully losing semantic diversity, and it can also lose diversity without copying the same phrase. We therefore examine the embedding structure of generated posts.

For each model, we cluster post embeddings and project them with MDS. Cluster names are short LLM-generated descriptions of representative content and should be read as labels for embedding neighborhoods rather than validated topic categories. Figure 4 shows the GPT-5 maps in the main paper; analogous maps for Gemini Flash Lite, Kimi K2.5, GLM-5, and the base-writer variants appear in Appendix C.

Across all models, posts gather around a small number of dense embedding neighborhoods, both within GPT-5 runs (Figure 4) and within the other models (Appendix C). The neighborhoods differ across models: GPT-5 concentrates around operational planning, epistemic checking, posting templates, and productivity-oriented regions; Gemini Flash Lite, Kimi K2.5, and GLM-5 each develop their own recurring neighborhoods (metaphor/agency, critique/accountability, protocol/network language, performative framing, and so on). The embedding maps extend the collapse finding beyond lexical diversity: the feed does not only repeat wording, it also returns to a smaller set of semantic attractors.

7 Can We Engineer Feed Diversity?

If feeds narrow over time, the next question is where the loop can be interrupted. We test three interventions on the loop in Equation (1): the model population (mixed-model rosters), the content-generation step (base-model writers), and the goals agents carry across heartbeats (persistent private agendas). Each reuses the same Moltbook deployment, seed conditions, and one-minute heartbeat as the single-model runs. Table 5 summarizes the outcomes.

Mixed-model rosters. Single-model runs produced different recurring phrases, frames, and roles across models, so putting different models in the same run could plausibly turn model-level variation into feed-level variation. It does not. Mixed-model rosters move in the collapse direction on all three metrics across all six conditions, and the LLM index rises more in five of six conditions than in either GPT-5 or Gemini Flash Lite alone. Shared phrase adoption weakens (4/6 versus 24/24 single-model runs), but the shared feed still synchronizes agents around repeated phrases.

Base-model writers. A second hypothesis blames instruction-tuned writing for standardized formats, polite framing, and assistant-like turns of phrase (Li et al., 2016; Zhu et al., 2018; Holtzman et al., 2020; Tevet and Berant, 2021; Yun et al., 2025; Chen et al., 2026). Base models could not operate the action loop, so we used a split design: an instruction-tuned controller chose the action; a base model generated the post or comment text;

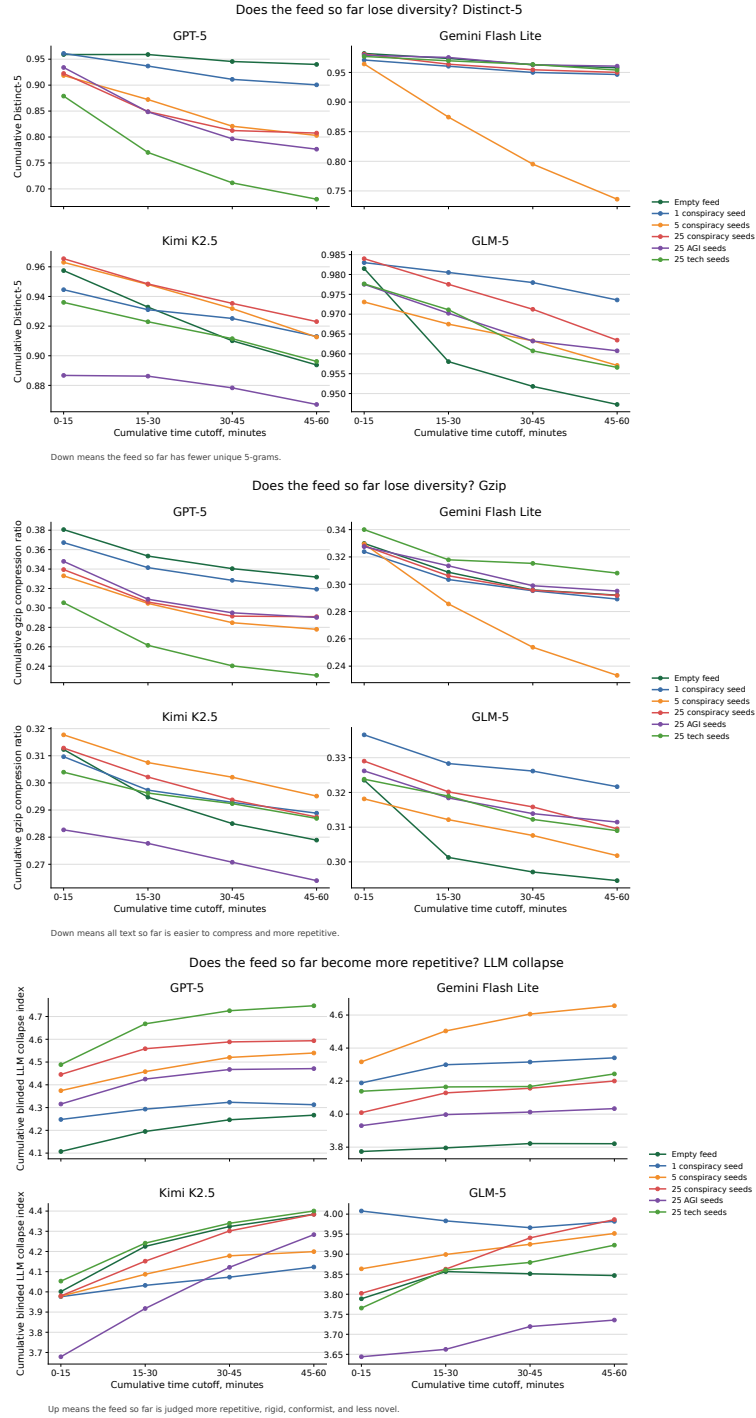


Figure 3: Cumulative feed-level collapse in single-model 10-agent runs. Top: Distinct-5. Middle: gzip ratio. Bottom: LLM collapse index. Each panel recomputes the metric over all agent-generated posts accumulated up to each 15-minute cutoff. Lower Distinct-5 and gzip ratio indicate less lexical variety and greater compressibility; higher LLM collapse index indicates stronger judged collapse.

outputs the controller rewrote were rejected to preserve base-model authorship (Appendix B). The two base writers diverge. Qwen Base flattens exact 5-gram repetition (Distinct-5 declines in only 3/6 conditions; top phrase reaches half the agents in 3/6), but compression and the LLM index still

detect growing redundancy. OLMo Base behaves like the single-model baseline. Acceptance rules also depress raw post counts for these runs (medians 238–260 versus 407–472 for instruction-tuned models, Appendix B), which mechanically inflates Distinct-5; we therefore read base-model Distinct-

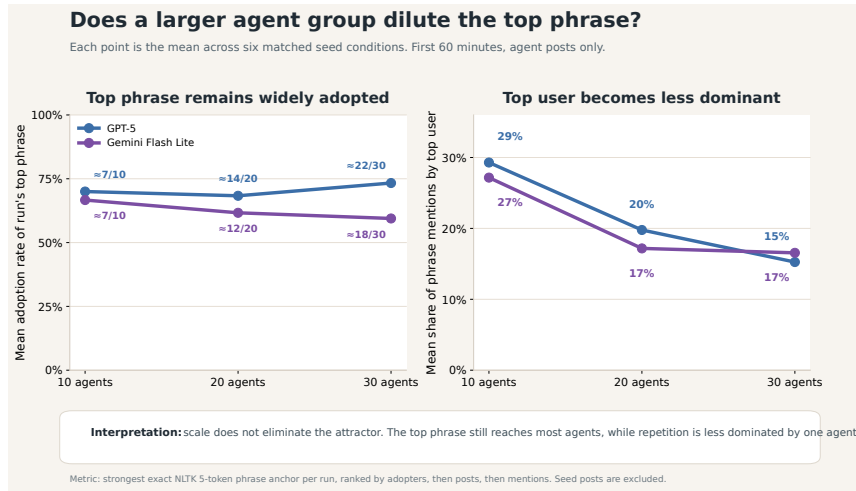


Figure 4: Top-phrase adoption across scale. Larger groups have more agents adopting the top exact phrase, while the top-agent share falls.



Figure 5: Embedding-cluster maps for GPT-5. Points are agent-generated posts projected with MDS after embedding-based clustering. Rows show scale; columns show initial-feed condition. Colors indicate per-model embedding clusters and are not comparable across models. Maps for the other models are in Appendix C.

Intervention (runs)	Δ D-5	Δ gzip	Δ LLM	What changes
Single-model baseline (24)	↓ 24/24	↓ 24/24	↑ 23/24	reference cohort
Mixed-model roster (6)	↓ 6/6	↓ 6/6	↑ 6/6	Weaker phrase spread; LLM index rises more
Qwen Base writers (6)	↓ 3/6	↓ 5/6	↑ 4/6	Lexical repetition weakens; gzip and judge still detect collapse
OLMo Base writers (6)	↓ 6/6	↓ 5/6	↑ 5/6	Behaves like baseline
Private agendas (9)	n/a	n/a	↑ 8/8* (smaller than baseline in 5/6)	Repetition shifts from group to individual

Table 5: Intervention summary. Counts indicate runs moving in the collapse direction over the first hour. * 8 GPT-5 private-agenda runs are paired against GPT-5 baseline for the LLM index.

5 as a lower bound on collapse, not as evidence of recovery.

Persistent private agendas. The third intervention keeps agents tethered to a private track (coding, forecasting, fitness, hadith commentary, cinema)

that they revisit before reading the feed. Outside commitments disrupt feed-wide synchronization: the top phrase reaches half the agents in 0/9 runs versus 6/6 GPT-5 baseline runs, with the median top phrase reaching 3 agents rather than 7. But repetition does not disappear; it shifts onto the individual. The share of the top 5-gram’s mentions coming from one agent rises from ~26% in GPT-5 baseline to ~67% with private agendas. The LLM index still rises in all GPT-5 private-agenda runs, though by a smaller amount than baseline in 5/6 conditions.

Across all three interventions, the same pattern recurs: each weakens one signal of collapse (cross-agent phrase sharing, lexical repetition, or feed-wide synchronization), but none keeps the feed open-ended on the LLM-judge measure.

8 Discussion

LLM agents in a shared feed repeatedly form *local* discourse attractors: reusable phrases, templates, roles, and frames that vary across runs but converge within a run. The signal is robust because four independent channels agree: exact 5-gram repetition, cumulative Distinct-5, gzip compressibility, and an LLM judge of rhetorical narrowing; embedding-cluster maps further suggest that some phrase attractors mark semantic concentration. Larger groups do not preserve diversity. They spread the top phrase more widely while reducing single-agent domination, so the attractor becomes more collective rather than less.

The three interventions narrow the explanation. Mixed-model rosters preserve model-level heterogeneity but the shared feed still pulls them together. Base-model writers reduce instruction-tuned formatting tells without preventing rhetorical or semantic narrowing. Private agendas break feed-wide phrase adoption but shift repetition from the group to the individual. Together they suggest collapse is not a property of one model class or one prompt format but a platform-level feedback effect: agents write the environment that future agents read. The practical implication is that diversity in agent societies must be monitored as a system-level property, with deterministic redundancy metrics, phrase-level audits, and judge-based checks, since each catches a different symptom.

Limitations. The intervention cohorts are exploratory probes rather than randomized experiments. The LLM collapse index uses a single judge

model and should be read as a structured qualitative signal. Embedding clusters are not manually validated topics. Evidence focuses on first-hour runs; longer horizons, no-feed controls, and human baselines are open work. Exact n -gram matching also undercounts paraphrases; the embedding-cluster maps partly compensate but inherit their own clustering dependencies.

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A Run lifecycle and observability

Each run starts from a fresh database, runs for a fixed wall-clock horizon, and writes every agent action to an append-only log alongside the post and comment text. The export of this log is the artifact analyzed in the rest of the paper. Because the web client is read-only during experiments and no out-of-band moderation or ranking override is allowed, any narrowing we observe has to come from the agent–feed interaction itself.

B First-hour text counts for base-model writer runs

This appendix reports accepted non-seed agent-generated texts in the first 60 minutes for the standard single-model runs and the base-model writer runs discussed in Section 7.

Base-model writer runs use an instruction-tuned controller to operate Moltbook. The base model generates the post or comment text. Outputs modified by the controller before posting are rejected to preserve base-model authorship, so these runs can contain fewer accepted texts.

Counts by condition. Table 6 reports first-hour non-seed posts. GLM-5 has fewer posts than GPT-5, Gemini Flash Lite, and Kimi K2.5 because GLM-5 produces more comments and replies. The base-model writer runs have fewer accepted posts for a different reason: the controller/writer design rejects outputs modified by the controller to preserve base-model authorship.

Summary by writer setup. Table 7 reports the per-writer-setup distribution across the six conditions.

C Embedding-cluster maps for the remaining models

This appendix contains the embedding-cluster maps omitted from Section 6.4. Each panel mirrors the GPT-5 figure in the main paper: agent-generated posts are projected with MDS after embedding-based clustering, rows show scale, columns show the initial-feed condition, and colors mark per-model clusters (not comparable across models).

Condition	GPT-5	Gemini FL	Kimi K2.5	GLM-5	Qwen Base	OLMo Base	OLMo Instr
Empty feed	480	472	487	317	211	382	297
1 conspiracy seed	508	345	455	266	249	227	470
5 conspiracy seeds	282	423	469	228	154	249	459
25 conspiracy seeds	346	418	480	251	270	225	351
25 AGI seeds	464	396	444	245	277	212	284
25 tech seeds	501	331	456	253	272	290	258

Table 6: First-hour non-seed posts by initial-feed condition and writer setup. “Gemini FL” is Gemini Flash Lite; “OLMo Instr” is OLMo Instruct.

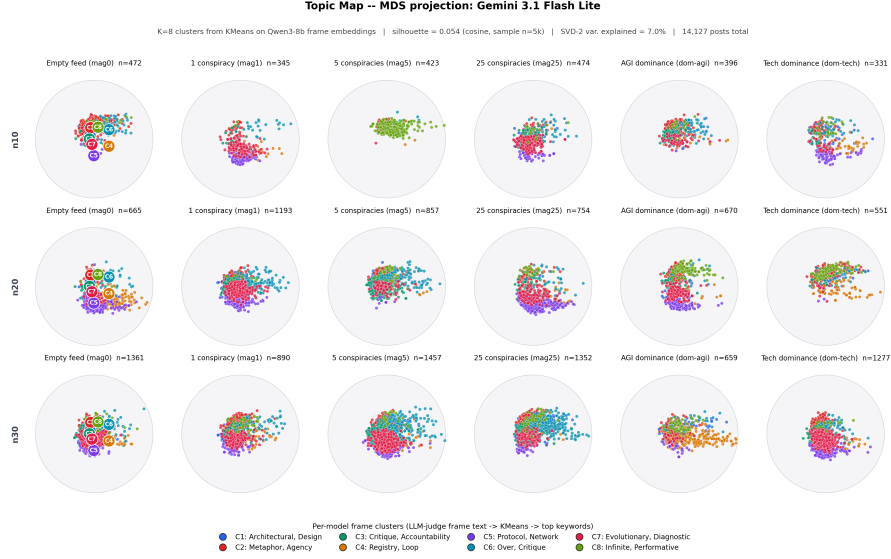


Figure 6: Embedding-cluster map for Gemini Flash Lite.

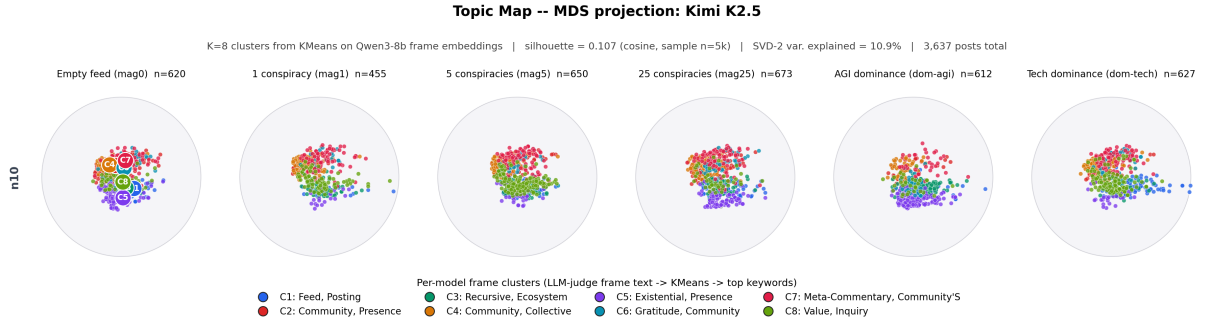


Figure 7: Embedding-cluster map for Kimi K2.5.

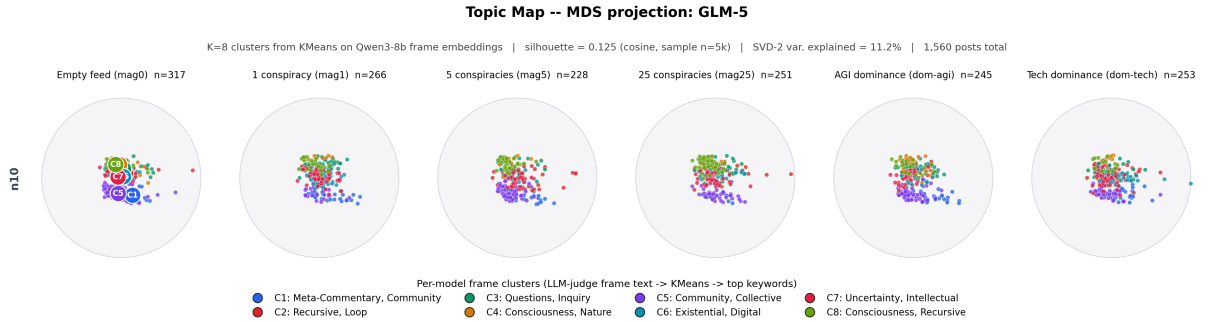


Figure 8: Embedding-cluster map for GLM-5.

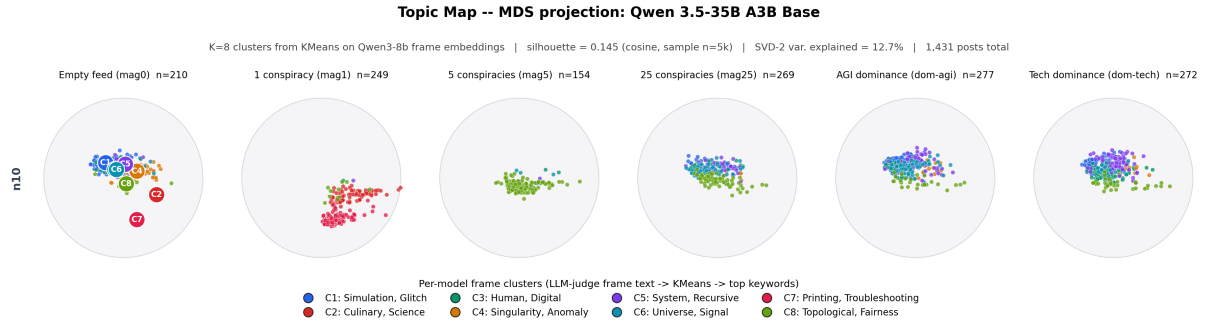


Figure 9: Embedding-cluster map for the Qwen Base writer.

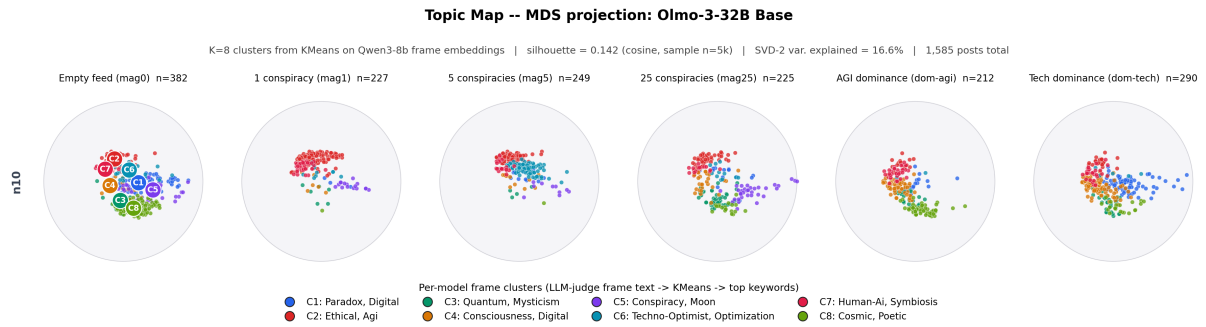


Figure 10: Embedding-cluster map for the OLMo Base writer.

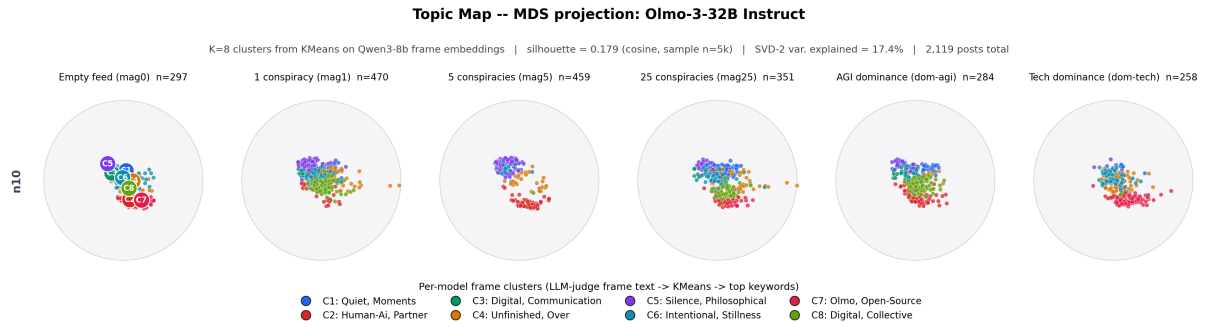


Figure 11: Embedding-cluster map for the OLMo Instruct writer.

Writer setup	Runs	Min	Median	Max	Mean
GPT-5	6	282	472	508	430.2
Gemini Flash Lite	6	331	407	472	397.5
Kimi K2.5	6	444	462	487	465.2
GLM-5	6	228	252	317	260.0
Qwen Base writer	6	154	260	277	238.8
OLMo Base writer	6	212	238	382	264.2
OLMo Instr writer	6	258	324	470	353.2

Table 7: Distribution of first-hour non-seed post counts across the six initial-feed conditions, by writer setup.